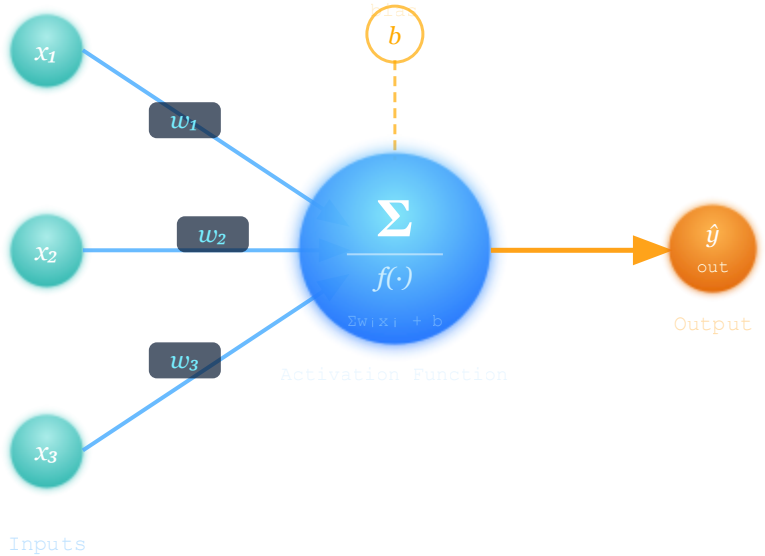
Artificial Neuron

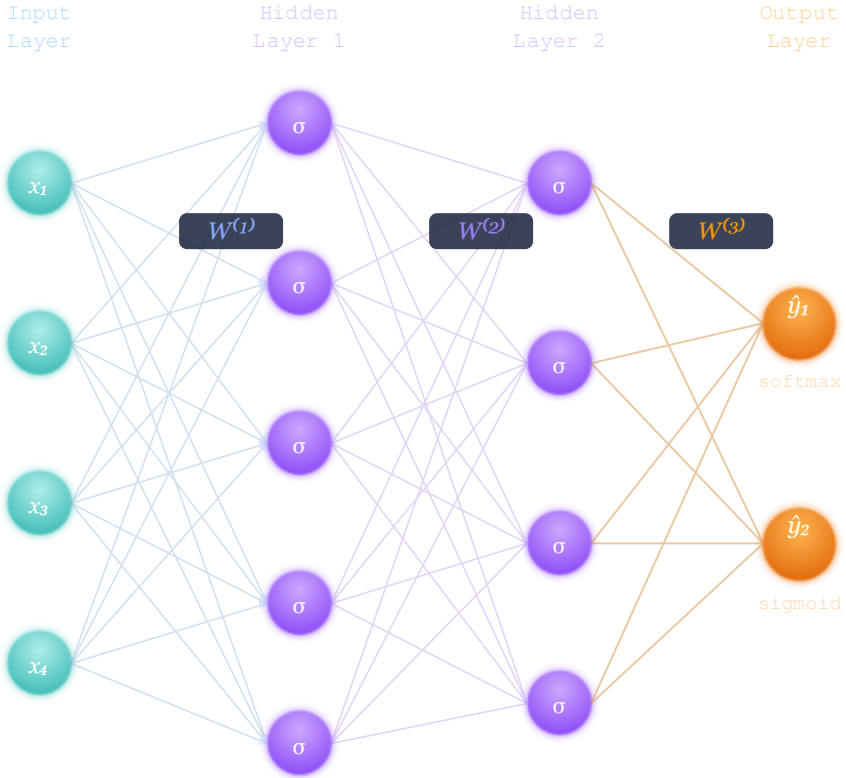


Weighted sum: $z = w_1x_1 + w_2x_2 + w_3x_3 + b$

Activation: $\hat{y} = f(z)$ e.g. sigmoid, ReLU, tanh

$f(z) = 1/(1+e^{-z})$ or $f(z) = \max(0, z)$

Neural Network



Each node = one artificial neuron (Σ + $f(\cdot)$)

Layers stacked: input → hidden(s) → output

Training: backpropagation adjusts weights W

[Generated by Claude Sonnet 4.6]